Practice Midterm 2
Introduction to Algorithms
601.433/633

Name:

Ethics Statement

I agree to complete this exam without unauthorized assistance from any person, materials, or device.

Signature: Date:

1 True/False (15 points)

Indicate whether each statement is true or false. You **do not** need to give a proof, nor do you need to give counter-examples.

(a) If I_1 and I_2 are independent sets in a matroid with $|I_1| > |I_2|$, then $I_2 \cup \{e\}$ is independent for all $e \in I_1 \setminus I_2$.
true false

(b) If all edge weights are distinct in a graph, then there is only one MST.
true false

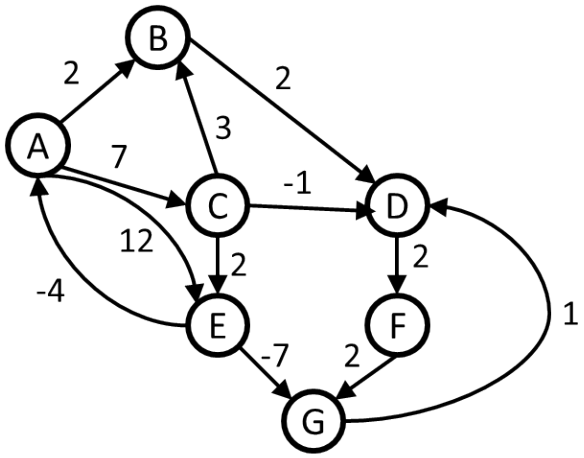
(c) In a bottom-up dynamic programming algorithm with a 2-dimensional table with n rows and m columns, we can always fill in the entries via two nested for loops of the form: for($i = 1$ to n) for($j = 1$ to m) $M[i, j] = \ldots$
true false

(d) Bellman-Ford gives correct distances even when there are negative edge weights, as long as there are no negative-weight cycles.
true false

(e) The node with smallest finishing time in a call to DFS(G) is in a source strongly connected component.
true false

2 Shortest Paths (30 points)

Consider the following graph.



Even though the graph has negative weight edges, let's consider what happens if we run Dijkstra's algorithm.

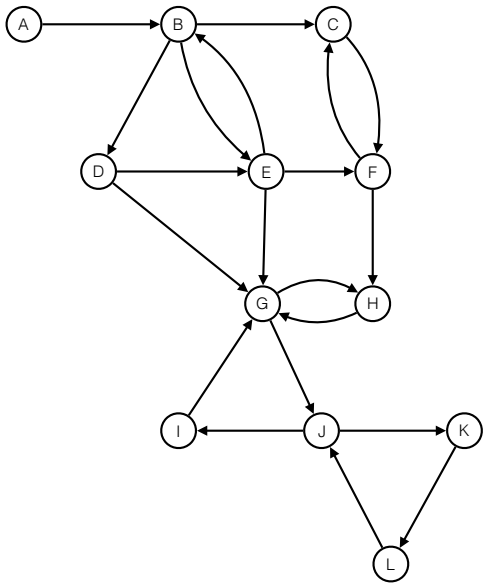
(a) (15 points) Let T be the shortest-path tree constructed by Dijkstra's algorithm on G (starting from A). List the vertices of G in the order in which they were added to T.

(b) (10 points) Draw the tree T computed by Dijkstra's algorithm on G (starting from A).

(c) (5 points) Dijkstra's algorithm found the wrong path to some nodes. For one of these nodes, write down the actual shortest path.

3 Strongly Connected Components (15 points)

List the strongly connected components of the following graph, and give a topological sort of the graph of the SCCs (i.e., the graph obtained by creating a vertex for each SCC and adding an edge from u to v if there is an edge in the given graph from the SCC that u represents to the SCC that v represents)



4 Kruskal's Algorithm (15 points)

Assume you have an implementation of the Union-Find data structure that has an amortized time of $f(n)$ for Union operations and amortized time of $g(n)$ for Find operations. If you use this data structure to implement Kruskal's algorithm on a connected graph with n nodes and m edges, what will be the running time of Kruskal's? Explain your answer.

5 Dynamic Programming (25 points)

Let $G = (V, E)$ be a directed graph with vertices $v_1, v_2, \ldots, v_n$. Recall that $|E| = m$. We say that G is an *ordered graph* if it has the following two properties:

1. Every edge goes from a vertex with a lower index to a vertex with a higher index. That is, every edge has the form (v_i, v_j) with $i < j$.

2. Every vertex other than v_n has at least one outgoing edge.

The length of a path is the number of edges on it. In this problem you will design an efficient algorithm to find the length of the *longest* path which starts at v_1 and ends at v_n. Note that by the two properties of an ordered graph, every node has at least one path to v_n.

(a) (10 points) Let $S(i)$ denote the length of the longest path that starts at v_i and ends at v_n. Write a (recursive) formula for $S(i)$, which in the next part will form the basis of a dynamic programming algorithm. You do not need a formal proof of correctness, but should justify your answer.

(b) (15 points) Design an $O(m)$-time algorithm to find the longest path which starts at v_1 and ends at v_n (using your solution from part (a)). Argue correctness and running time (again, you do not need a fully formal proof, but should provide justification).